



AI-DRIVEN

CYBER SECURITY

*AI DOESN'T SEARCH DATABASES LIKE HUMANS DO
IT SEARCHES MEANING, AND TODAY YOU'LL LEARN HOW.*

TRADITIONAL SQL DATABASE

- STORES STRUCTURED DATA
- USES TABLES AND ROWS
- SEARCHES EXACT VALUES
- USES SQL QUERIES
- GREAT FOR STRUCTURED INFORMATION

VECTOR DATABASE

- STORES NUMERICAL REPRESENTATIONS
- SEARCHES BY MEANING
- FINDS SIMILAR CONTENT
- USES VECTOR SIMILARITY
- USEFUL FOR AI APPLICATIONS

SQL VS VECTOR DATABASE

- **SQL:** "FIND THIS EXACT VALUE"
- **VECTOR:** "FIND SOMETHING SIMILAR"
- SQL → STRUCTURED DATA
- VECTOR DB → SEMANTIC SEARCH
- THEY CAN WORK TOGETHER

EMBEDDINGS

- TEXT CONVERTED INTO NUMBERS
- CAPTURES MEANING
- SIMILAR MEANING → SIMILAR VECTORS
- USED FOR SEMANTIC SEARCH
- GENERATED BY EMBEDDING MODELS

CHUNKING

- SPLITS LARGE DOCUMENTS INTO SMALLER PIECES
- MAKES INFORMATION EASIER TO SEARCH
- EACH CHUNK GETS AN EMBEDDING
- IMPROVES RETRIEVAL
- IMPORTANT FOR RAG

SIMILARITY SEARCH

- COMPARES VECTORS
- FINDS MOST RELEVANT CONTENT
- USES DISTANCE/SIMILARITY
- **COMMON METHODS:** COSINE SIMILARITY
- RETURNS TOP MATCHING RESULTS

CHROMADB

- OPEN-SOURCE VECTOR DATABASE
- PYTHON-FRIENDLY
- STORES EMBEDDINGS + METADATA
- SUPPORTS SIMILARITY SEARCH
- GREAT FOR RAG PROTOTYPES

PERSISTENT VECTOR INDEX

- SAVES VECTORS TO DISK
- DATA SURVIVES PROGRAM RESTART
- AVOIDS RE-EMBEDDING EVERYTHING
- ENABLES REUSABLE AI KNOWLEDGE BASES